# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
## 6th Semester of B. Pharm Examination
## University Theory Examination April-May 2015
### PHARMACEUTICAL BIOCHEMISTRY-II (PH311)

**Date: 04.05.2015, Monday** **Time: 10.00 a.m to 1.00 p.m** **Maximum Marks: 80**

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**Instructions:**
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

### SECTION- I

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|---|---|---|---|
| **Q 1** | Attempt any **TWO** of the followings | | |
| A | Write a detailed note on *de novo* pathway for the biosynthesis of purine nucleotides with structural representation. | | **10** |
| B | a) | Give an account on possible enzyme alteration causing Gout. | **04** |
| | b) | Discuss biosynthesis of porphyrin with structural representation. | **06** |
| C | a) | What do you mean by $N_2$ cycle? Give an account on $N_2$ fixation by atmosphere. | **05** |
| | b) | Write a note on sulphur cycle with suitable examples of sulphur containing amino acids in the body. | **05** |
| **Q 2** | Attempt any **TWO** of the followings | | |
| A | Describe in details DNA replication with all essential enzymes. Explain the significances of DNA replication. | | **10** |
| B | Write a note on following: | | **10** |
| | a) | Transcription process | |
| | b) | Disorders of porphyrin biosynthesis | |
| C | Answer the following questions: | | **10** |
| | a) | Give an account on nitrification process involved in $N_2$ Cycle with suitable examples of nitrifying bacteria. | |
| | b) | Give chemical structure of purine and pyrimidine bases. | |

## SECTION- II

**Q 3** Attempt any **TWO** of the following

A. Explain the chemiosmotic theory of oxidative phosphorylation. **10**

B. Classify vitamins and add a detailed note on metabolism and function of vitamine D. **10**

C.
a) Name various oxidoreductase enzymes and explain biological role of any two of them. **05**
b) What are vitamins? Explain importance of vitamin E and K. **05**

**Q 4** Attempt any **TWO** of the following

A. Classify liver function tests and explain various tests used for estimation of bilirubin in serum. **10**

B. Add a short note on following: **10**

a) PCR
b) Immune fluorescence

C. Define electrophoresis. Classify electrophoretic techniques and explain them in detail. **10**